Shlok Mehndiratta

BS in Electrical Engineering and Computer Science, IISER Bhopal

+91-8218635012 | shlokm23@iiserb.ac.in | LinkedIn | GitHub | Portfolio

Education

Indian Institute of Science Education and Research Bhopal (IISER Bhopal) — Bhopal, India
Bachelor of Science in Electrical Engineering and Computer Science (EECS) — *Aug 2023 – Present*

- CPI: **8.87/10 (3.55/4)**

Central Board of Secondary Education (CBSE) — Roorkee, India
Grade XII: 95.8% | Grade X: 96.8% — *2021 – 2023*

Research Experience

Vision-Based Autonomous Navigation and Swarm Coordination — *MOON Lab, IISER Bhopal*
Undergraduate Research Student | Advisor: Dr. P. B. Sujit — *May 2026 – Present*

- Implementing spherical coordinate-based vision planning with extensions for moving target tracking and dynamic obstacle avoidance for MAVs, validating algorithms on ROS 2 simulation environment.
- Concurrently assembling drone systems integrating flight controller, RC receiver, and telemetry modules; configuring and integrating OptiTrack mocap system with ROS for high-precision indoor localization during algorithm testing and validation.
- Progressing toward multi-agent coordination: investigating decentralized swarm algorithms for cooperative coverage and autonomous missions.

Learning Visual Contact Estimation from Photorealistic Synthetic Data — *RRC, IIIT Hyderabad*
Research Intern | Advisor: Dr. Spandan Roy — *April 2026 – Present*

- Building high-fidelity synthetic data engine generating paired multi-view RGB images, segmentation masks, and 3D contact/SDF supervision using GPU-accelerated physics (NVIDIA Warp) and photorealistic rendering (BlenderProc).
- Training vision-conditioned neural models (ResNet/UNet/PointNet) to predict 3D contact distributions from camera images, leveraging synthetic supervision for downstream grasp generation and real-robot policy transfer on 16-DoF LEAP Hand platforms.
- Targeting paper submission on learning-based contact prediction from visual data to top-tier robotics venues; scaling dataset generation across multi-GPU clusters for large-scale synthetic scene diversity.

Perception-Aware Altitude Optimization for Aerial-Ground Tracking — *IISER Bhopal*
Undergraduate Research Student | Advisor: Dr. Arijit Sen — *Sept 2025 – July 2026*

- Developed closed-form altitude scheduling policy for UAV-UGV cooperative tracking by formalizing the altitude-coverage-resolution coupling; built belief-space MPC controller and event-triggered aerial planner achieving 25× compute reduction while maintaining estimation quality.
- Constructed complete ROS 2/Gazebo testbed with unified logging protocol to validate theoretical predictions on sensor noise scaling, altitude optimality, and event-triggered performance across variable terrain curvature.

Publications

Equi-partitioning coverage achieved via a hybrid interaction rule by distributed team of Unmanned Vehicles.
Santosh Tantravahi, Nirman Jaiswal, **Shlok Mehndiratta**, P.B. Sujit, Aditya Paranjape

Developed decentralized algorithm enabling autonomous drone swarms to dynamically partition and monitor large areas; validated across 35+ real-world flight experiments using 3–5 Crazyflie nano-quadcopters and integrated wildfire detection framework with ROS 2/SITL simulations. [**Submitted to ICRA 2027**]

COAST: Altitude as a Coupled Perception Variable in Shared-Camera Aerial-Ground Tracking
Shlok Mehndiratta[†], Divyam Sood[†], Prashant Gupta[†], Agniva Banerjee, Arijit Sen

Formal write-up of the altitude-coverage-accuracy coupling and closed-form policy described above, with full derivation and reproducible ROS 2/Gazebo validation. [**Submitted to ACODS 2026**]

Open Source Contributions

ros2_control: ROS 2 Control Framework

PRs | *Issues*

- Improved real-time safety and reliability: migrated controller manager lock to RT-safe priority-inheritance mutex to prevent timing delays in the control loop; implemented dynamic command toggling for testing hardware communication failures.
- Enhanced debugging and testing: added comprehensive exception-handling tests ensuring plugin failures do not cascade; improved error messages to identify which hardware component failed, reducing diagnosis time.
- Modernized framework: refactored deprecated parameter APIs across the ecosystem, eliminating technical debt and preventing future API breakage.

Technical Skills

Programming Languages: Python, C++, MATLAB, Verilog, Assembly
Robotics & Simulation: ROS2, Gazebo, NVIDIA Isaac Sim, Model Predictive Control, Extended Kalman Filter
Computer Vision & ML: PyTorch, TensorFlow, OpenCV, SLAM, Reinforcement Learning
Data & ML Libraries: NumPy, Pandas, scikit-learn
Systems & Deployment: Git/GitHub, Linux/Bash, Docker, FastAPI, LaTeX

Additional Projects

AI-Powered Contract Review & Negotiation Assistant | *Infosys Springboard Virtual Internship* [GitHub]

- Built end-to-end LLM-based contract analysis application with FastAPI backend, Flutter cross-platform frontend, and Docker containerization; developed full-stack AI solution for legal document processing and automated negotiation assistance.

CS50's Web Programming with Python and JavaScript | *Harvard University* [Project Demos]

- Built full-stack web applications including a single-page email client, a Twitter-like social network, and an eBay-style auction platform using Django, JavaScript, and SQL.

Co-Curricular Activities

Computing and Networking Council, IISER Bhopal Aug 2024 – April 2026

Vice Secretary (Aug 2024 – Sept 2025), Technical Head for Techno-Cult Festival 2025

- Founded EECS Club and organized 10+ workshops on ROS, ML, and embedded systems; managed infrastructure and technical events for 1,500+ participant techno-cultural festival.

Google Developer Group, IISER Bhopal Aug 2024 – Aug 2025

Web Development Lead

- Directed campus platform development and mentored students in software engineering practices and collaborative development.

Student Placement Representative IISER Bhopal

EECS Department

Interests: Lawn Tennis, Cricket, Guitar

Research Interests

Autonomous robotics and intelligent systems. Combining classical control with learning methods for real-world deployment. Multi-modal perception, robot control, and multi-agent coordination.